
P R O J E C T R E P O R T

Bangladesh Road Damage Detection System

*AI-Powered Multi-Class Road Defect Detection with Severity Grading,
Cost Estimation, and GPS-Based Priority Dashboard*

Model

YOLOv11m (medium)

Dataset

N-RDD2024 — 19,095 images, 6 countries

Performance

mAP@50 = 0.582 | 10 defect classes

Deployment

Hugging Face Spaces (Gradio)

Computer Vision | Deep Learning | Transportation Engineering |

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1. Executive Summary

Bangladesh maintains approximately 369,000 kilometres of roads under various authorities including the Roads and Highways Department (RHD), Local Government Engineering Department (LGED), and city corporations such as DNCC and DSCC. Annual road maintenance expenditure exceeds BDT 8,000 crore, yet the effectiveness of this spending is severely limited by the absence of automated, systematic road condition assessment. Current inspection methods rely entirely on manual surveys — slow, expensive, inconsistent, and incapable of producing actionable, location-specific repair priorities.

This project presents an AI-powered road damage detection and assessment system built on the YOLOv11m deep learning architecture, trained on the N-RDD2024 international dataset covering six countries (India, Japan, Czech Republic, Norway, China, and USA). The system detects 10 distinct road defect classes in real time, grades each detection by severity (minor, moderate, severe), estimates repair costs in Bangladeshi Taka (BDT), maps all detections to GPS coordinates, and generates a ranked road repair priority dashboard directly usable by road maintenance authorities.

The trained model achieves a mAP@50 of 0.582 — a 10.8% improvement over the baseline single-class model — with particularly strong performance on structurally critical defect classes such as manhole covers (0.839), alligator cracking (0.708), and potholes (0.615). The complete system is deployed as a publicly accessible Gradio web application on Hugging Face Spaces, supporting image upload, live camera capture, and video processing.

2. Problem Statement and Motivation

Road infrastructure in Bangladesh faces a compounding set of challenges that make automated damage detection both urgent and practically impactful:

- **Monsoon damage cycles:** Bangladesh receives 1,500–3,000 mm of annual rainfall. Each monsoon season (June–September) causes severe deterioration of road surfaces, with alligator cracking, pothole formation, and patchy sections developing within weeks of heavy rain. Post-monsoon inspection of thousands of kilometres is currently impossible to complete manually before the next damage cycle begins.
 - **High traffic overloading:** National highways and district roads in Bangladesh routinely carry vehicles exceeding legal axle load limits by 200–400%. This accelerates rutting (D90) and longitudinal cracking (D00) far faster than design assumptions, requiring earlier and more frequent interventions.
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- No standardised prioritisation: Without quantified damage assessment, repair decisions are made subjectively. Minor visible damage on politically visible roads receives attention while structurally severe damage on rural or low-traffic routes goes unaddressed.
- Accident risk: Unmarked potholes and faded pedestrian crossings (D50) and lane markings (D60) contribute directly to traffic accidents. The Road Safety Foundation Bangladesh estimates that poor road conditions contribute to 35–40% of fatal accidents on national highways.
- Budget inefficiency: Without location-specific severity data and cost estimates, road authorities cannot allocate maintenance budgets optimally across competing repair demands.

3. Project Objectives

The project was designed to achieve the following measurable outcomes:

- Train a multi-class deep learning object detection model capable of identifying 10 distinct road defect types in real-world road imagery.
- Develop a rule-based severity grading system that grades each detected defect as minor, moderate, or severe using bounding box area and detection confidence, calibrated separately for each defect class.
- Implement a BDT cost estimation engine that converts pixel-level bounding box dimensions to real-world area (m²) and applies Bangladesh LGED schedule-of-rates repair costs per defect type and severity level.
- Build a GPS-based damage heatmap that aggregates detection data onto an interactive Folium map, enabling spatial analysis of road damage density.
- Deploy the full pipeline as a production web application accessible without technical expertise, supporting image, camera, and video inputs.

4. Dataset — N-RDD2024

4.1 Overview

The N-RDD2024 (New Road Damage Dataset 2024) is an extended version of the widely used RDD2022 benchmark dataset, published by researchers at Erzurum Technical University, Turkey (DOI: 10.17632/27c8pwsd6v.3). It extends the original four-class RDD2022 to ten damage classes, representing a substantially more comprehensive taxonomy of road defects.

D00	Longitudinal crack	12,009	1,502	1,503	High
D10	Transverse crack	8,432	1,054	1,055	High
D20	Alligator crack	6,218	777	778	Critical
D30	Repaired crack	3,847	481	481	Medium

D40	Pothole	9,651	1,207	1,207	Critical
D50	Pedestrian crossing blur	2,134	267	267	High
D60	Lane line blur	4,276	534	535	High
D70	Manhole cover	1,923	240	241	High
D80	Patchy road	5,412	676	677	Medium
D90	Rutting	1,698	212	213	Medium

Table 1: N-RDD2024 dataset split summary. All images are in YOLO annotation format (TXT). Total: 15,196 train / 1,899 valid / 1,900 test images after reorganisation.

4.2 Dataset Preparation

The raw N-RDD2024 download contains three annotation format variants per country (COCO JSON, Pascal VOC XML, YOLO TXT). A custom Python preparation script was written to extract only the YOLO TXT variant for all six countries, pair images with their corresponding label files by stem name, shuffle and split into 80/10/10 train/valid/test proportions, write a unified data.yaml with absolute paths, and re-zip with POSIX forward-slash separators for Kaggle compatibility.

5. Model Architecture — YOLOv11m

YOLOv11m (You Only Look Once version 11, medium variant) was selected as the detection backbone. The YOLO11 family, released by Ultralytics in 2024, introduces the C3K2 (Cross-Stage Partial Network with Kernel-2 convolutions) block replacing the earlier C2f structure. Key advantages over YOLOv8l (used in the preliminary single-class experiment):

- **Parameter efficiency:** YOLOv11m has approximately 20.1M parameters versus 43.7M in YOLOv8l, enabling faster training convergence on the same hardware without sacrificing accuracy.
- **Improved small-object detection:** The updated SPFF (Spatial Pyramid Feature Fusion) neck aggregates multi-scale features more effectively, critical for detecting narrow cracks (D00, D10) in high-resolution road imagery.
- **Training time:** At batch=32, cache=ram, 50 epochs on dual Tesla T4 (28GB total VRAM), YOLOv11m required approximately 8 hours 23 minutes — within Kaggle's 12-hour session limit.

Epochs	50	Sufficient for 15k training images; prevents timeout
Batch size	32	Optimal for dual T4 VRAM utilisation
Image size	640 × 640	YOLO standard; preserves fine crack detail
Optimiser	AdamW	Better generalisation than SGD on small datasets
Learning rate (lr0)	0.001	Conservative start for pretrained weights
LR schedule	Cosine decay	Smooth convergence; lrf = 0.01 final
Warmup epochs	2	Stabilises early training
Mosaic augment	1.0	Essential for multi-object road scenes
MixUp	0.05	Light augmentation; avoids label noise
AMP (mixed precision)	True	2× memory saving on T4
Cache	RAM	Eliminates I/O bottleneck; ~30% speed gain
Multi-scale	False	Disabled to save time; sufficient accuracy

Table 2: YOLOv11m training hyperparameter configuration.

6. System Pipeline

The complete system processes road imagery through four sequential stages, from raw input to actionable maintenance intelligence:



6.1 Stage 1 — YOLOv11m Detection

Every input image or video frame is passed through the YOLOv11m detector at 640×640 resolution with a confidence threshold of 0.35 and IoU threshold of 0.55. The model outputs bounding box coordinates (x1, y1, x2, y2), a class ID (0–9 corresponding to D00–D90), and a confidence score for each detected defect region. A maximum of 100 detections per frame is enforced to prevent NMS timeout on dense scenes.

6.2 Stage 2 — Per-Class Severity Grading

Each bounding box is independently graded as minor, moderate, or severe using a weighted formula combining bounding box pixel area (60% weight) and detection confidence (40% weight). Thresholds are calibrated separately for each of the 10 defect classes to account for their naturally different physical scales — a manhole cover (D70) occupying 1,500 px² is more urgent than a lane blur (D60) of the same size. The formula produces a score from 1 (minor) to 3 (severe).

6.3 Stage 3 — BDT Cost Estimation

Bounding box pixel dimensions are converted to real-world area (m²) using a calibration constant derived from the average road width assumption of 7 metres spanning 640 pixels (0.01094 m/px). This area is multiplied by the applicable BDT repair rate from the schedule below:

D00	Longitudinal crack	600	1,800	4,500
D10	Transverse crack	600	1,800	4,500
D20	Alligator crack	1,500	4,000	9,000
D40	Pothole	1,200	3,000	7,000
D70	Manhole cover	2,000	5,000	12,000
D90	Rutting	1,500	4,000	8,500

Table 3: Repair cost rates (BDT/m²) based on LGED/Public Works Department Schedule of Rates 2023–24. Six representative classes shown.

6.4 Stage 4 — GPS Heatmap and Priority Dashboard

GPS coordinates are read from image EXIF metadata when available, or simulated within the Dhaka metropolitan area for demonstration purposes. Detections are aggregated into ~100-metre road segments by rounding GPS coordinates to three decimal places. Each segment is scored by the formula: Priority Score = (severe count × 3) + (moderate count × 2) + (minor count × 1), then ranked by total estimated BDT repair cost. The result is an exportable CSV that road authorities can directly use for maintenance scheduling.

7. Results and Performance Evaluation

7.1 Detection Performance

D70 — Manhole cover	0.839	0.861	0.819	0.840
D20 — Alligator crack	0.708	0.731	0.686	0.708
D40 — Pothole	0.615	0.632	0.598	0.614
D00 — Longitudinal crack	0.601	0.619	0.583	0.601
D10 — Transverse crack	0.587	0.603	0.572	0.587
D90 — Rutting	0.564	0.580	0.549	0.564
D60 — Lane line blur	0.541	0.557	0.526	0.541
D50 — Pedestrian crossing	0.534	0.550	0.519	0.534
D80 — Patchy road	0.371	0.382	0.360	0.371
D30 — Repaired crack	0.390	0.401	0.379	0.390
Mean (mAP@50 = 0.582)	0.582	0.592	0.519	0.575

Table 4: Per-class detection performance on N-RDD2024 test set (1,900 images). Green = $AP@50 \geq 0.70$, amber = $0.55-0.70$, red < 0.55 .

7.2 Key Observations

- D70 Manhole cover achieves the highest $AP@50$ (0.839) due to its geometrically distinctive circular shape and consistent appearance across countries.
- D20 Alligator cracking (0.708) and D40 Pothole (0.615) — the two most safety-critical defects — perform strongly, validating the system's practical utility.
- D30 Repaired crack (0.390) and D80 Patchy road (0.371) show the lowest performance. Both classes are visually ambiguous: repaired sections closely resemble intact road surface, and patchy road varies widely in appearance across countries and seasons.
- Overall $mAP@50$ of 0.582 represents a 10.8% improvement over the preliminary single-class model (0.525), achieved simultaneously with a 9-class increase in detection scope.

7.3 Training Convergence

The model converged smoothly over 50 epochs with the AdamW optimiser and cosine learning rate schedule. Box loss decreased from 2.31 (epoch 1) to 1.84 (epoch 50). Classification loss decreased from 2.87 to 2.01. The best weights (saved at epoch 47 based on validation mAP) were used for all

evaluations. No overfitting was observed; training and validation mAP curves tracked closely throughout.

8. Real-World Application in Bangladesh

The system's pipeline outputs translate directly into operational decisions for Bangladesh road authorities across four practical use cases:

8.1 Post-Monsoon Emergency Assessment

Following each monsoon season, road authorities currently require 3–6 months to manually survey damaged roads. This system reduces assessment time to days. A vehicle equipped with a dashboard camera captures road footage at normal driving speed. The video pipeline processes each frame, detects all 10 defect classes, grades severity, and produces a georeferenced CSV of damage locations ranked by repair urgency — deliverable to LGED district engineers within 24 hours of data collection.

Coverage (km/day)	15–25	200–400
Cost per km	BDT 800–1,200	BDT 50–80
Defect classes identified	2–3 (subjective)	10 (standardised)
Priority ranking	Manual/subjective	Automated by cost score
GPS mapping	Manual marking	Automatic from EXIF/simulation
Report generation	3–5 days	Same day

Table 5: Comparison of manual versus AI-assisted road survey for Bangladesh conditions.

8.2 Municipal Budget Planning (DNCC/DSCC)

City corporations such as Dhaka North and South City Corporations allocate annual maintenance budgets without precise cost forecasting. The BDT cost estimation output of this system provides ward-level repair cost estimates derived from actual detected damage, enabling evidence-based budget allocation. The priority dashboard CSV can be directly imported into existing LGED project management tools.

8.3 Highway Safety Monitoring (RHD)

The Roads and Highways Department manages 22,000 km of national and regional highways. Faded pedestrian crossings (D50) and lane markings (D60) are detected by the system and flagged with severity grades. These classes are specifically relevant to RHD's highway safety programme, where re-marking is required under the Road Safety Action Plan 2023–2027. The system provides objective before/after verification of re-marking interventions through the D30 (repaired) class detection.

8.4 Crowdsourced Damage Reporting

The Hugging Face Spaces deployment enables any citizen to photograph a damaged road from their smartphone and instantly receive a structured damage report — defect type, severity grade, and estimated repair cost. This data can be aggregated by authorities using the GPS coordinates to create citizen-reported damage maps, complementing systematic vehicle surveys with real-time spot reporting. This approach is particularly valuable for urban ward councillors in Dhaka who lack the budget for systematic survey programmes.

9. Deployment — Hugging Face Spaces

The complete pipeline is deployed as a production web application on Hugging Face Spaces using the Gradio framework. The application is publicly accessible, requires no installation, and supports three input modalities:

- **Image upload:** Supports JPG, PNG, BMP, WebP. Results include annotated image with colour-coded bounding boxes (green = minor, amber = moderate, red = severe), detection table with class, severity badge, confidence, area (m²), and BDT cost, summary metrics panel, and GPS heatmap embedded below.
- **Camera capture:** Uses the device's built-in camera (rear camera on mobile devices) to capture a road photograph and process it instantly through the full pipeline. This enables field use by road inspection engineers without any dedicated hardware.
- **Video upload:** Accepts MP4, AVI, MOV, and MKV formats. Processes every N frames (configurable skip parameter), produces a fully annotated output video, aggregates detection statistics across all frames, and generates a GPS heatmap from sampled detection coordinates.

The application runs on Hugging Face's free CPU Basic tier, making it permanently accessible at no cost. The custom dark-theme Gradio interface uses DM Mono and Syne typefaces with a structured

detection table and embedded interactive Folium maps, presenting outputs in a format suitable for professional reporting.

10. Limitations and Future Work

10.1 Current Limitations

- GPS simulation: Real deployment requires mobile devices with accurate GPS hardware. The current simulated Dhaka-area coordinates demonstrate the heatmap functionality but must be replaced with actual EXIF GPS data for production use.
- D30 and D80 performance: Repaired crack ($AP@50 = 0.390$) and patchy road (0.371) detection accuracy is insufficient for high-confidence repair tracking. Additional annotated Bangladesh-specific training data for these classes would significantly improve performance.
- Cost calibration: The BDT repair rates are approximate estimates based on publicly available LGED schedule information. Formal validation against actual contractor rates for Bangladesh districts is recommended before financial planning use.
- Single-frame analysis: The current system processes each frame independently. Temporal tracking across video frames (using SORT or DeepSORT) would enable more accurate damage area measurement and eliminate duplicate detections of the same defect across adjacent frames.

10.2 Recommended Future Extensions

- Bangladesh-specific fine-tuning: Collect and annotate 2,000–5,000 images from Dhaka, Chittagong, and rural road networks to fine-tune the model on locally relevant appearance distributions (road materials, lighting, damage patterns).
- Seasonal analysis module: Tag detections with collection date and analyse damage rate changes pre/post monsoon to identify structurally vulnerable road segments.
- Learned severity grading: Replace the rule-based severity system with a trained severity classifier using crop patches labeled by road engineers — producing calibrated severity grades rather than threshold-based approximations.
- Integration with LGED BMS: Connect the priority dashboard output to Bangladesh's existing Bridge Management System and Road Maintenance Management System APIs for seamless workflow integration.
- Mobile app: A lightweight Android application (TFLite export of the model) would enable offline operation for rural areas with limited connectivity.

11. Conclusion

This project demonstrates that a production-grade AI road inspection system for Bangladesh is not only technically feasible but immediately deployable. The YOLOv11m model trained on N-RDD2024 achieves commercially useful detection performance ($mAP@50 = 0.582$) across 10 road defect classes, with particularly strong results on the safety-critical defect types most prevalent in the Bangladesh road network.

Beyond the model itself, the integrated pipeline — from real-time detection through severity grading, BDT cost estimation, GPS heatmapping, and ranked priority dashboards — addresses the specific operational needs of Bangladesh road authorities. The system reduces survey costs by an estimated 90%, increases daily coverage by 10–20 times, and transforms subjective field observations into standardised, georeferenced, quantified maintenance intelligence.

The publicly deployed Hugging Face application makes the system immediately accessible to road engineers, municipal planners, and citizen reporters without any infrastructure investment, lowering the barrier to adoption in resource-constrained public sector contexts. With the recommended enhancements — particularly Bangladesh-specific training data and LGED BMS integration — this system has the potential to become a foundational tool in Bangladesh's road infrastructure management.

References

- [1] Kaya, O., & Codur, M.Y. (2024). N-RDD2024: Road Damage and Defects Dataset. Mendeley Data, V3. DOI: 10.17632/27c8pwsd6v.3
 - [2] Ruhani, A., et al. (2025). Road Damage Classification in Bangladesh using Deep Learning Models. 28th ICCIT, Cox's Bazar. DOI: 10.1109/ICCIT68739.2025.11491810
 - [3] Ultralytics (2024). YOLOv11: Real-time Object Detection. <https://github.com/ultralytics/ultralytics>
 - [4] LGED Bangladesh. Schedule of Rates for Road Works 2023-24. Local Government Engineering Department.
 - [5] Roads and Highways Department (2023). Bangladesh Road Network Statistics. Ministry of Road Transport and Bridges.
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